

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA0401

COURSE OUTCOME COVERED

CO1: *Analyse the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)*

QUESTIONS: 05

Total Marks:100

Annexure A

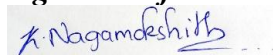
Exploratory Data Analysis - Based Practical Assessment

Code of Conduct:

I Nagamokshith Koduru (192424083) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Exploratory Data Analysis-Based Practical Assessment

1. Education Analytics Dataset

Question 1

A college wants to analyze student performance based on attendance, study hours, internal marks, and final result. Perform EDA to identify the factors affecting student results.

Student_ID	Attendance	Study_Hours	Internal_Marks	Result
S01	92	5	45	Pass
S02	55	2	24	Fail
S03	78	4	38	Pass
S04	60	3	28	Fail
S05	88	5	42	Pass
S06	47	1	20	Fail
S07	95	6	48	Pass
S08	72	3	35	Pass
S09	50	2	22	Fail
S10	83	4	40	Pass
S11	66	3	30	Fail
S12	90	5	44	Pass
S13	58	2	25	Fail
S14	76	4	36	Pass
S15	62	3	29	Fail
S16	85	5	41	Pass
S17	45	1	18	Fail
S18	80	4	39	Pass
S19	68	3	32	Pass
S20	53	2	23	Fail

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# Load the dataset
file_path = '/content/Student_Performance.xlsx'
df = pd.read_excel(file_path)
# Display the first 5 rows of the DataFrame
print('First 5 rows of the dataset:')
display(df.head())
# Display basic information about the dataset
print('\nInformation about the dataset:')
df.info()
# Display descriptive statistics
print('\nDescriptive statistics of the dataset:')
display(df.describe())
```

```

# Check for missing values
print('\nMissing values in each column:')
display(df.isnull().sum())
# Convert 'Result' column to numerical representation if it's not already.
# Assuming 'Pass' is 1 and 'Fail' is 0 for numerical analysis.
# First, check the unique values and their type.
if df['Result'].dtype == 'object':
    # Map 'Pass' to 1 and 'Fail' to 0
    df['Result_Numeric'] = df['Result'].map({'Pass': 1, 'Fail': 0})
    numerical_features_with_result = ['Attendance', 'Study_Hours', 'Internal_Marks',
'Result_Numeric']
else:
    numerical_features_with_result = ['Attendance', 'Study_Hours', 'Internal_Marks', 'Result']
plt.figure(figsize=(15, 10))
for i, feature in enumerate(numerical_features_with_result):
    plt.subplot(2, 2, i + 1)
    sns.histplot(df[feature], kde=True)
    plt.title(f'Distribution of {feature}')
    plt.xlabel(feature)
    plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
plt.figure(figsize=(18, 6))

# Bar plot for average Attendance by Result
plt.subplot(1, 3, 1)
sns.barplot(x='Result', y='Attendance', data=df, palette='viridis')
plt.title('Average Attendance by Result')
plt.xlabel('Result')
plt.ylabel('Average Attendance')

# Bar plot for average Study Hours by Result
plt.subplot(1, 3, 2)
sns.barplot(x='Result', y='Study_Hours', data=df, palette='plasma')
plt.title('Average Study Hours by Result')
plt.xlabel('Result')
plt.ylabel('Average Study Hours')

# Bar plot for average Internal Marks by Result
plt.subplot(1, 3, 3)
sns.barplot(x='Result', y='Internal_Marks', data=df, palette='magma')
plt.title('Average Internal Marks by Result')
plt.xlabel('Result')
plt.ylabel('Average Internal Marks')

plt.tight_layout()
plt.show()

```

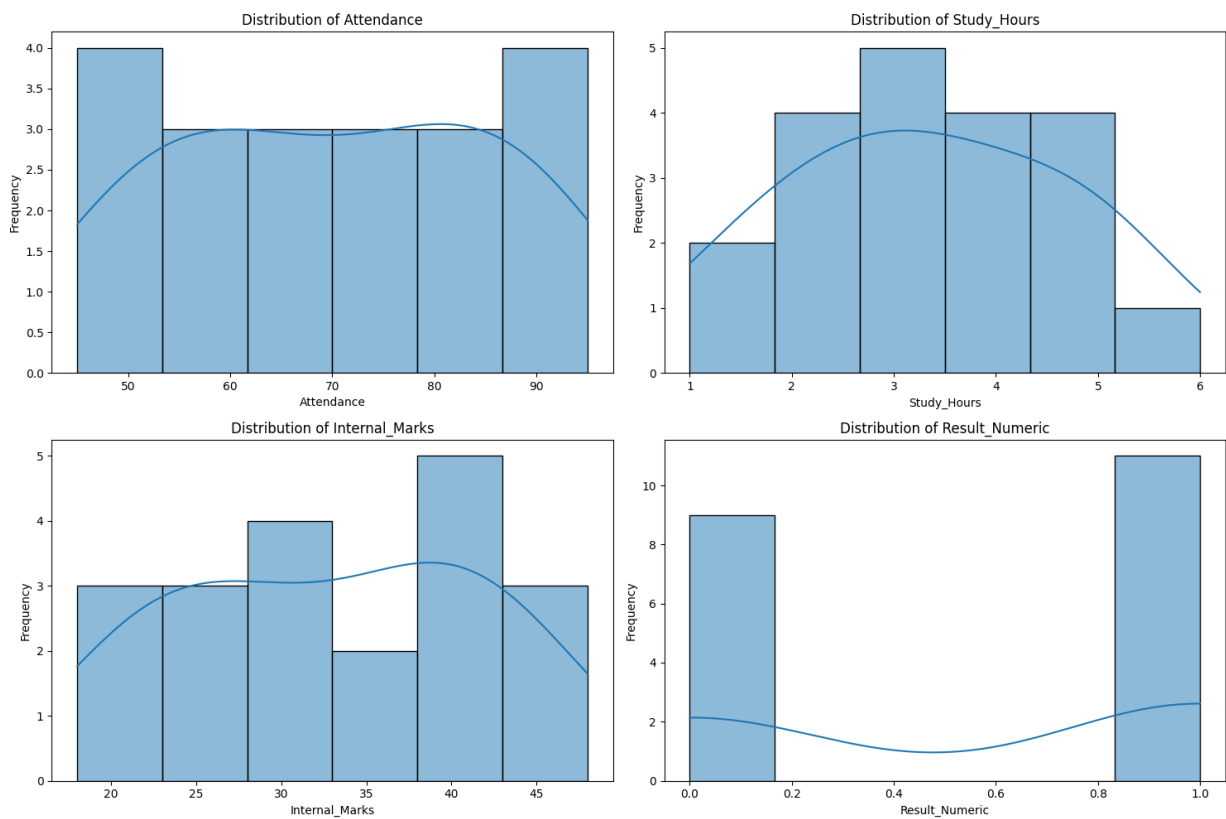
OUTPUT:

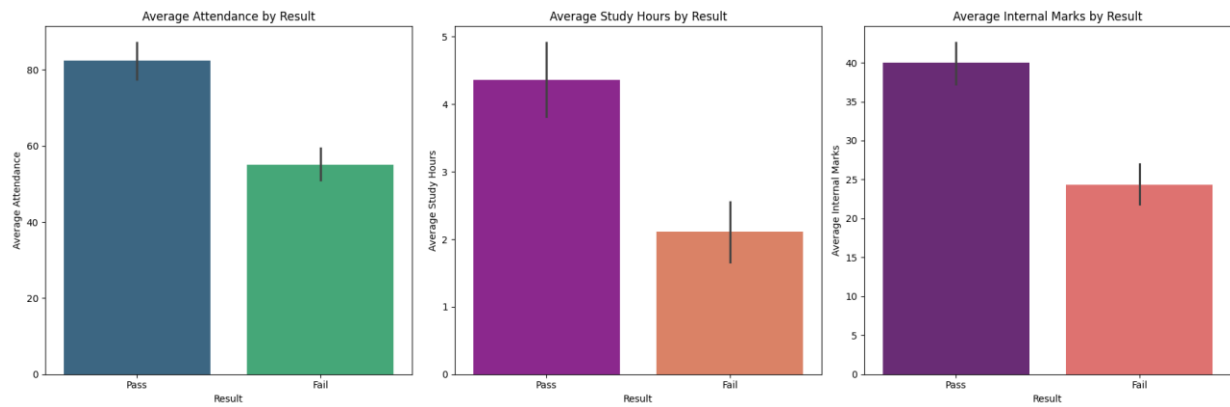
First 5 rows of the dataset:

	Student_ID	Attendance	Study_Hours	Internal_Marks	Result	Unnamed: 5	Unnamed: 6	Unnamed: 7
0	S01	92	5	45	Pass	NaN	NaN	NaN
1	S02	55	2	24	Fail	NaN	NaN	NaN
2	S03	78	4	38	Pass	NaN	NaN	NaN
3	S04	60	3	28	Fail	NaN	NaN	NaN
4	S05	88	5	42	Pass	NaN	NaN	NaN

Descriptive statistics of the dataset:

	Attendance	Study_Hours	Internal_Marks	Unnamed: 5	Unnamed: 6	Unnamed: 7
count	20.000000	20.000000	20.000000	0.0	0.0	1.0
mean	70.150000	3.350000	32.950000	NaN	NaN	5.0
std	15.938286	1.424411	9.087151	NaN	NaN	NaN
min	45.000000	1.000000	18.000000	NaN	NaN	5.0
25%	57.250000	2.000000	24.750000	NaN	NaN	5.0
50%	70.000000	3.000000	33.500000	NaN	NaN	5.0
75%	83.500000	4.250000	40.250000	NaN	NaN	5.0
max	95.000000	6.000000	48.000000	NaN	NaN	5.0





2. Healthcare Analytics Dataset

Question 2

A hospital wants to analyze patient health risk based on age, blood sugar, blood pressure, BMI, and disease status. Perform EDA to find important health patterns.

Patient_ID	Age	Sugar_Level	BP	BMI	Disease_Status
P01	45	145	130	28	Yes
P02	32	98	118	23	No
P03	55	180	145	31	Yes
P04	28	90	110	22	No
P05	60	200	150	34	Yes
P06	40	120	125	26	No
P07	52	165	140	30	Yes
P08	35	105	120	24	No
P09	48	155	135	29	Yes
P10	30	95	115	23	No
P11	62	210	155	35	Yes
P12	38	115	122	25	No
P13	50	170	142	30	Yes
P14	27	88	108	21	No
P15	58	190	148	33	Yes
P16	42	130	128	27	No
P17	46	150	132	28	Yes
P18	34	100	116	24	No
P19	53	175	144	31	Yes
P20	29	92	112	22	No

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# Load the Healthcare Analysis dataset
file_path_healthcare = '/content/Healthcare_Analysis.xlsx'
df_healthcare = pd.read_excel(file_path_healthcare)

# Display the first 5 rows of the DataFrame
print('First 5 rows of the Healthcare Analysis dataset:')
```

```

display(df_healthcare.head())
# Display basic information about the healthcare dataset
print("\nInformation about the Healthcare Analysis dataset:")
df_healthcare.info()
# Display descriptive statistics for the healthcare dataset
print("\nDescriptive statistics of the Healthcare Analysis dataset:")
display(df_healthcare.describe())
# Check for missing values in the healthcare dataset
print("\nMissing values in each column of the Healthcare Analysis dataset:")
display(df_healthcare.isnull().sum())
numerical_features_healthcare = ['Age', 'Sugar_Level', 'BP', 'BMI']

plt.figure(figsize=(15, 10))
for i, feature in enumerate(numerical_features_healthcare):
    plt.subplot(2, 2, i + 1)
    sns.histplot(df_healthcare[feature], kde=True)
    plt.title(f'Distribution of {feature}')
    plt.xlabel(feature)
    plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
plt.figure(figsize=(18, 6))

# Bar plot for average Age by Disease Status
plt.subplot(1, 4, 1)
sns.barplot(x='Disease_Status', y='Age', data=df_healthcare, palette='viridis')
plt.title('Average Age by Disease Status')
plt.xlabel('Disease Status')
plt.ylabel('Average Age')

# Bar plot for average Blood Sugar by Disease Status
plt.subplot(1, 4, 2)
sns.barplot(x='Disease_Status', y='Sugar_Level', data=df_healthcare, palette='plasma')
plt.title('Average Blood Sugar by Disease Status')
plt.xlabel('Disease Status')
plt.ylabel('Average Blood Sugar')

# Bar plot for average Blood Pressure by Disease Status
plt.subplot(1, 4, 3)
sns.barplot(x='Disease_Status', y='BP', data=df_healthcare, palette='magma')
plt.title('Average Blood Pressure by Disease Status')
plt.xlabel('Disease Status')
plt.ylabel('Average Blood Pressure')

# Bar plot for average BMI by Disease Status
plt.subplot(1, 4, 4)
sns.barplot(x='Disease_Status', y='BMI', data=df_healthcare, palette='cividis')
plt.title('Average BMI by Disease Status')
plt.xlabel('Disease Status')
plt.ylabel('Average BMI')

```

```
plt.tight_layout()
plt.show()
```

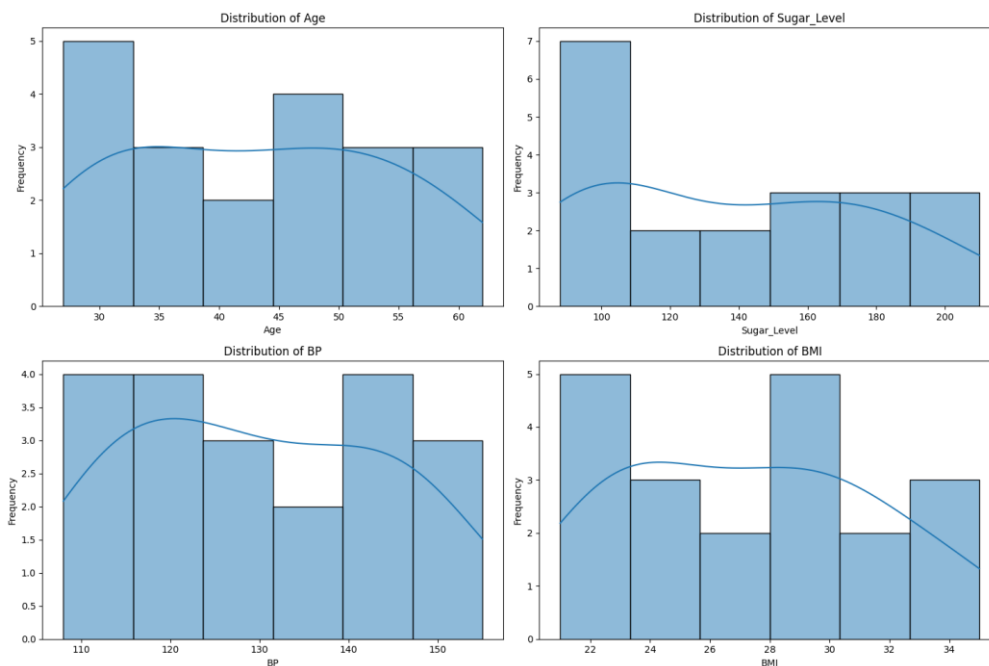
OUTPUT:

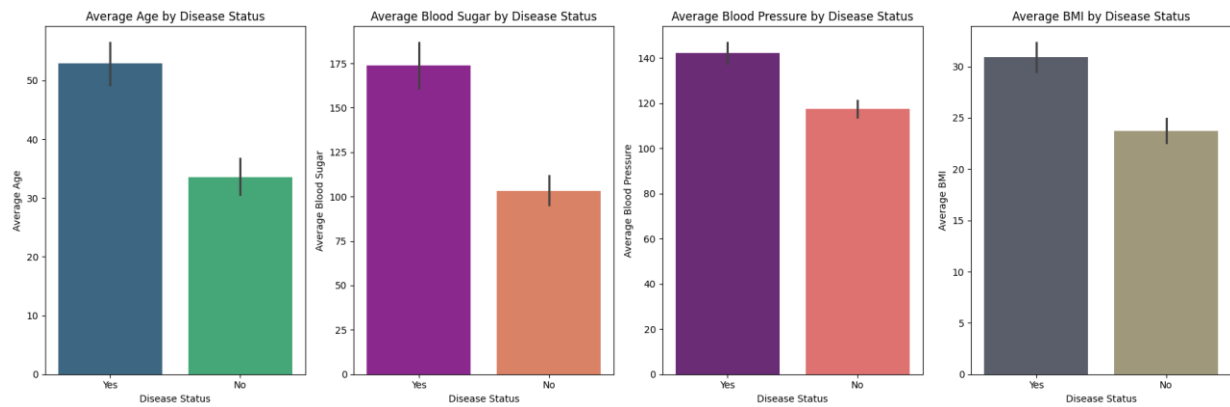
First 5 rows of the Healthcare Analysis dataset:

	Patient_ID	Age	Sugar_Level	BP	BMI	Disease_Status
0	P01	45	145	130	28	Yes
1	P02	32	98	118	23	No
2	P03	55	180	145	31	Yes
3	P04	28	90	110	22	No
4	P05	60	200	150	34	Yes

Descriptive statistics of the Healthcare Analysis dataset:

	Age	Sugar_Level	BP	BMI
count	20.000000	20.000000	20.000000	20.000000
mean	43.200000	138.650000	129.750000	27.300000
std	11.316499	40.33449	14.523574	4.256265
min	27.000000	88.00000	108.000000	21.000000
25%	33.500000	99.50000	117.500000	23.750000
50%	43.500000	137.50000	129.000000	27.500000
75%	52.250000	171.25000	142.500000	30.250000
max	62.000000	210.00000	155.000000	35.000000





3. Retail Sales Dataset

Question 3

A supermarket wants to analyze product sales based on category, price, quantity sold, discount, and revenue. Perform EDA to identify high-selling products and revenue patterns.

Product ID	Category	Price	Quantity Sold	Discount	Revenue
PR01	Grocery	50	20	5	1000
PR02	Snacks	30	35	3	1050
PR03	Dairy	45	18	2	810
PR04	Fruits	60	25	5	1500
PR05	Vegetables	40	30	4	1200
PR06	Grocery	80	15	6	1200
PR07	Snacks	25	40	3	1000
PR08	Dairy	55	16	2	880
PR09	Fruits	70	22	5	1540
PR10	Vegetables	35	28	4	980
PR11	Grocery	90	12	6	1080
PR12	Snacks	20	45	2	900
PR13	Dairy	65	14	3	910
PR14	Fruits	75	20	5	1500
PR15	Vegetables	45	26	4	1170
PR16	Grocery	100	10	7	1000
PR17	Snacks	35	32	3	1120
PR18	Dairy	50	19	2	950
PR19	Fruits	80	18	6	1440
PR20	Vegetables	30	35	3	1050

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# Load the Retail Sales dataset
file_path_retail = '/content/Retail_Sales.xlsx'
df_retail = pd.read_excel(file_path_retail)
```

```
# Display the first 5 rows of the DataFrame
```



```

print('First 5 rows of the Retail Sales dataset:')
display(df_retail.head())
# Display basic information about the retail dataset
print('\nInformation about the Retail Sales dataset:')
df_retail.info()
# Display descriptive statistics for the retail dataset
print('\nDescriptive statistics of the Retail Sales dataset:')
display(df_retail.describe())
# Check for missing values in the retail dataset
print('\nMissing values in each column of the Retail Sales dataset:')
display(df_retail.isnull().sum())
numerical_features_retail = ['Price', 'Quantity_Sold', 'Discount', 'Revenue']

plt.figure(figsize=(15, 10))
for i, feature in enumerate(numerical_features_retail):
    plt.subplot(2, 2, i + 1)
    sns.histplot(df_retail[feature], kde=True)
    plt.title(f'Distribution of {feature}')
    plt.xlabel(feature)
    plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
plt.figure(figsize=(18, 6))

# Bar plot for Total Revenue by Category
plt.subplot(1, 2, 1)
sns.barplot(x='Category', y='Revenue',
data=df_retail.groupby('Category')['Revenue'].sum().reset_index(), palette='viridis')
plt.title('Total Revenue by Product Category')
plt.xlabel('Product Category')
plt.ylabel('Total Revenue')
plt.xticks(rotation=45, ha='right')

# Bar plot for Total Quantity Sold by Category
plt.subplot(1, 2, 2)
sns.barplot(x='Category', y='Quantity_Sold',
data=df_retail.groupby('Category')['Quantity_Sold'].sum().reset_index(), palette='plasma')
plt.title('Total Quantity Sold by Product Category')
plt.xlabel('Product Category')
plt.ylabel('Total Quantity Sold')
plt.xticks(rotation=45, ha='right')

plt.tight_layout()
plt.show()

```

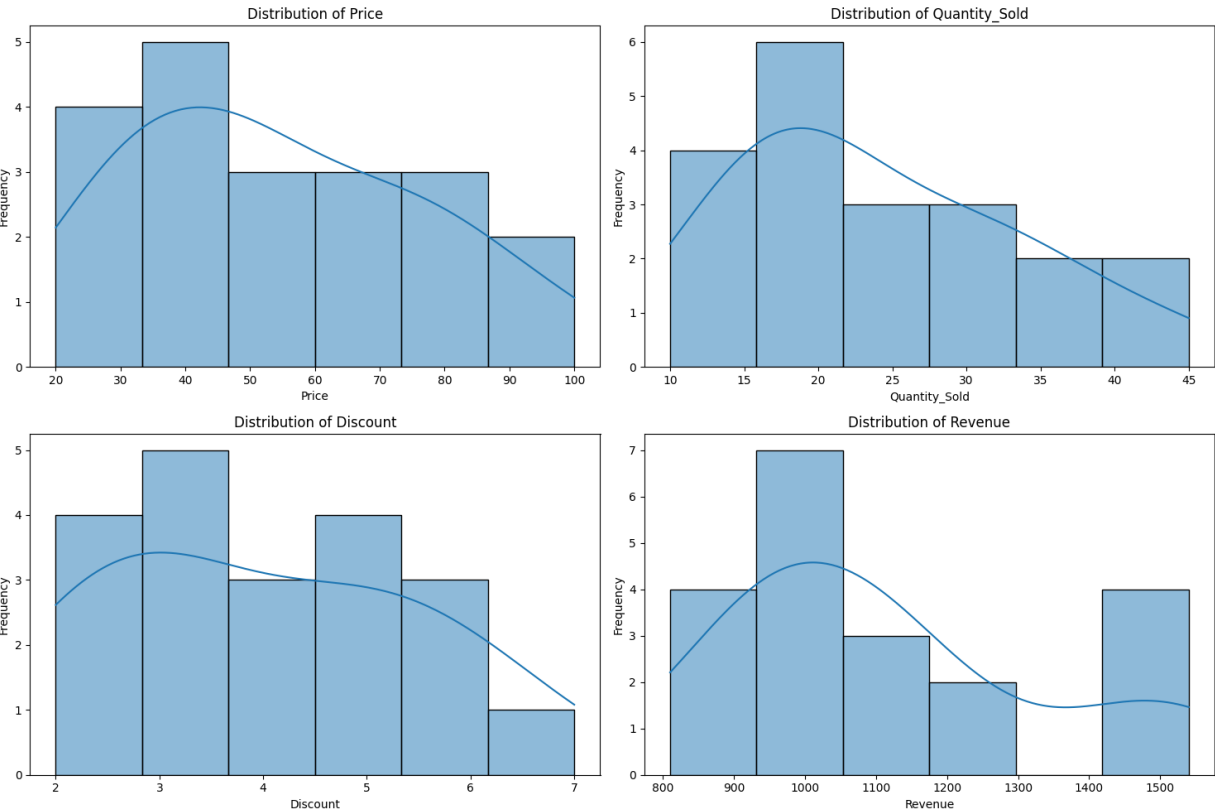
OUTPUT:

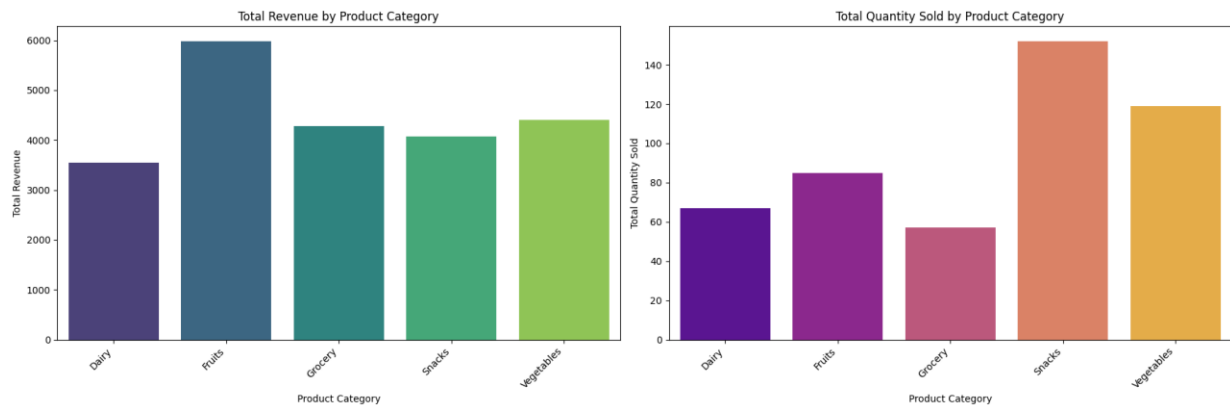
First 5 rows of the Retail Sales dataset:

	Product_ID	Category	Price	Quantity_Sold	Discount	Revenue
0	PR01	Grocery	50	20	5	1000
1	PR02	Snacks	30	35	3	1050
2	PR03	Dairy	45	18	2	810
3	PR04	Fruits	60	25	5	1500
4	PR05	Vegetables	40	30	4	1200

Descriptive statistics of the Retail Sales dataset:

	Price	Quantity_Sold	Discount	Revenue
count	20.000000	20.000000	20.000000	20.000000
mean	54.000000	24.000000	4.000000	1114.000000
std	22.803509	9.673621	1.555973	221.416493
min	20.000000	10.000000	2.000000	810.000000
25%	35.000000	17.500000	3.000000	972.500000
50%	50.000000	21.000000	4.000000	1050.000000
75%	71.250000	30.500000	5.000000	1200.000000
max	100.000000	45.000000	7.000000	1540.000000





4. Banking Loan Approval Dataset

Question 4

A bank wants to analyze loan approval decisions based on income, credit score, loan amount, employment type, and approval status. Perform EDA to identify patterns related to loan approval.

Customer ID	Income	Credit Score	Loan Amount	Employment Type	Loan Status
C01	45000	720	200000	Salaried	Approved
C02	25000	580	150000	Self-employed	Rejected
C03	60000	750	300000	Salaried	Approved
C04	22000	560	120000	Unemployed	Rejected
C05	50000	710	250000	Salaried	Approved
C06	30000	600	180000	Self-employed	Rejected
C07	70000	780	350000	Salaried	Approved
C08	28000	590	160000	Self-employed	Rejected
C09	55000	730	270000	Salaried	Approved
C10	24000	570	130000	Unemployed	Rejected
C11	65000	760	320000	Salaried	Approved
C12	35000	620	200000	Self-employed	Rejected
C13	48000	700	240000	Salaried	Approved
C14	26000	585	140000	Unemployed	Rejected
C15	75000	790	400000	Salaried	Approved
C16	32000	610	190000	Self-employed	Rejected
C17	52000	725	260000	Salaried	Approved
C18	27000	575	150000	Self-employed	Rejected
C19	68000	770	330000	Salaried	Approved
C20	23000	555	125000	Unemployed	Rejected

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# Load the Banking Loan Approval dataset
file_path_loan = '/content/Banking_Loan_Approval.xlsx'
df_loan = pd.read_excel(file_path_loan)
```

```
# Display the first 5 rows of the DataFrame
```

```

print('First 5 rows of the Banking Loan Approval dataset:')
display(df_loan.head())
# Display basic information about the loan dataset
print('\nInformation about the Banking Loan Approval dataset:')
df_loan.info()
# Display descriptive statistics for the loan dataset
print('\nDescriptive statistics of the Banking Loan Approval dataset:')
display(df_loan.describe())
# Check for missing values in the loan dataset
print('\nMissing values in each column of the Banking Loan Approval dataset:')
display(df_loan.isnull().sum())
numerical_features_loan = ['Income', 'Credit_Score', 'Loan_Amount']

plt.figure(figsize=(15, 5))
for i, feature in enumerate(numerical_features_loan):
    plt.subplot(1, 3, i + 1)
    sns.histplot(df_loan[feature], kde=True)
    plt.title(f'Distribution of {feature}')
    plt.xlabel(feature)
    plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
plt.figure(figsize=(18, 6))

# Bar plot for average Income by Approval Status
plt.subplot(1, 3, 1)
sns.barplot(x='Loan_Status', y='Income', data=df_loan, palette='viridis')
plt.title('Average Income by Approval Status')
plt.xlabel('Approval Status')
plt.ylabel('Average Income')

# Bar plot for average Credit Score by Approval Status
plt.subplot(1, 3, 2)
sns.barplot(x='Loan_Status', y='Credit_Score', data=df_loan, palette='plasma')
plt.title('Average Credit Score by Approval Status')
plt.xlabel('Approval Status')
plt.ylabel('Average Credit Score')

# Bar plot for average Loan Amount by Approval Status
plt.subplot(1, 3, 3)
sns.barplot(x='Loan_Status', y='Loan_Amount', data=df_loan, palette='magma')
plt.title('Average Loan Amount by Approval Status')
plt.xlabel('Approval Status')
plt.ylabel('Average Loan Amount')

plt.tight_layout()
plt.show()
plt.figure(figsize=(8, 5))
sns.countplot(x='Employment_Type', hue='Loan_Status', data=df_loan, palette='crest')
plt.title('Loan Approval Status by Employment Type')

```

```
plt.xlabel('Employment Type')
plt.ylabel('Count')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

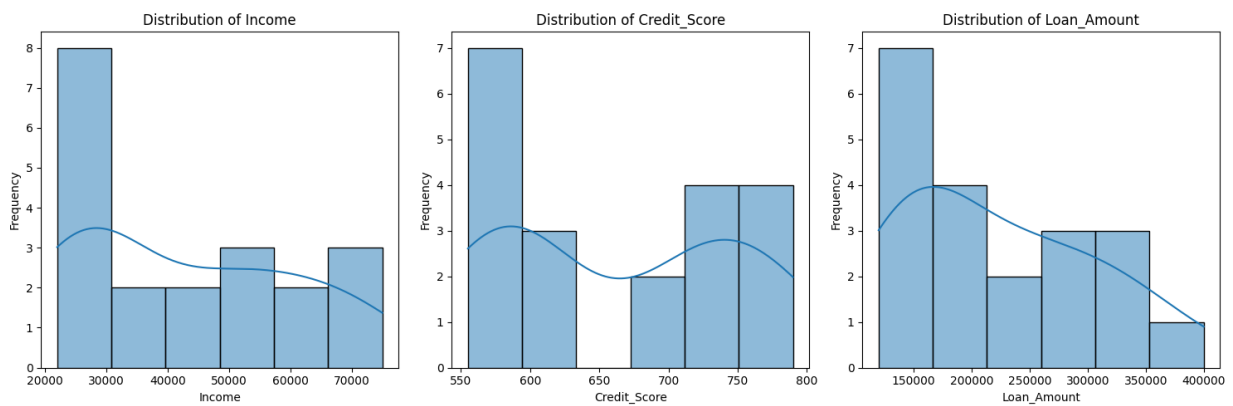
OUTPUT:

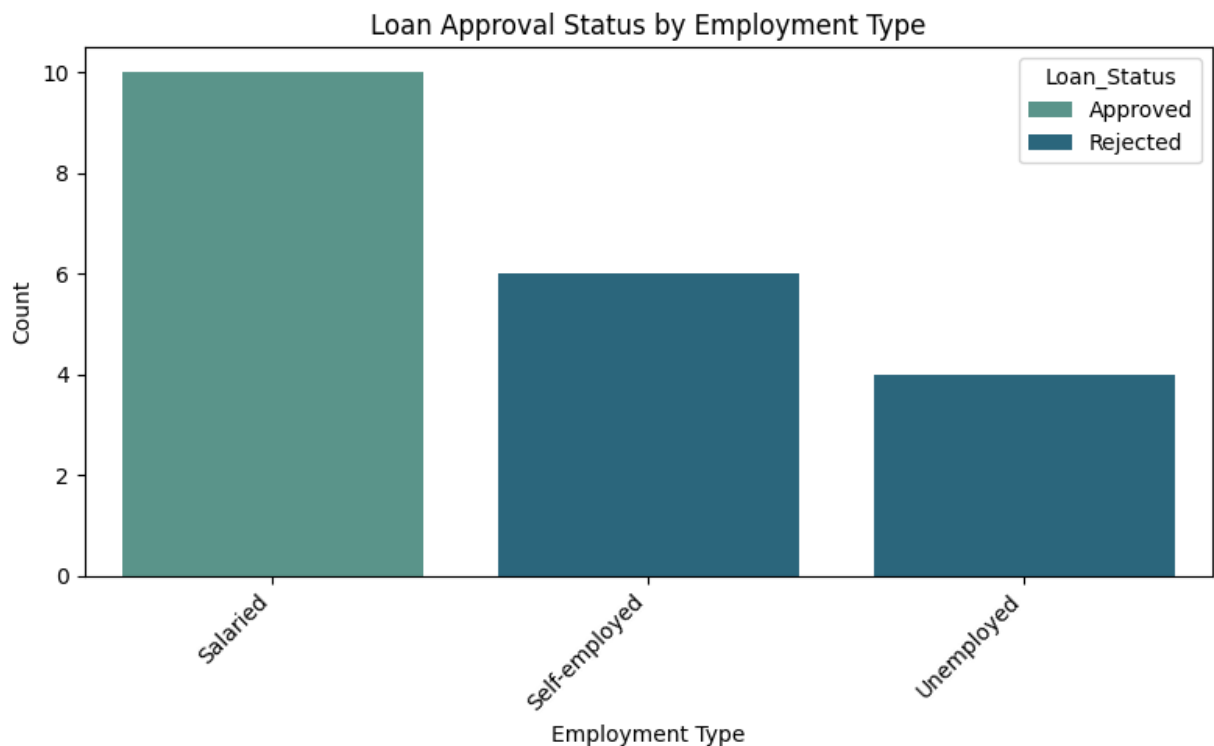
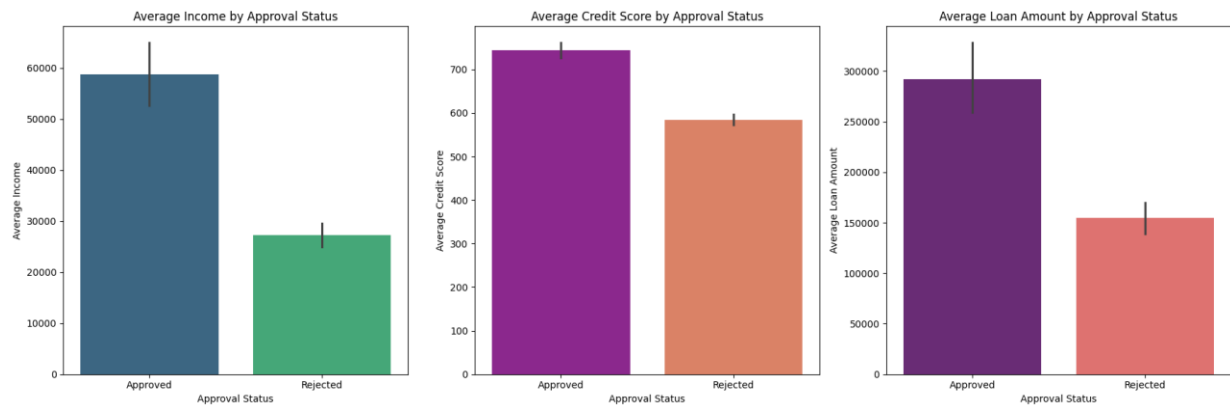
First 5 rows of the Banking Loan Approval dataset:

	Customer_ID	Income	Credit_Score	Loan_Amount	Employment_Type	Loan_Status
0	C01	45000	720	200000	Salaried	Approved
1	C02	25000	580	150000	Self-employed	Rejected
2	C03	60000	750	300000	Salaried	Approved
3	C04	22000	560	120000	Unemployed	Rejected
4	C05	50000	710	250000	Salaried	Approved

Descriptive statistics of the Banking Loan Approval dataset:

	Income	Credit_Score	Loan_Amount
count	20.000000	20.000000	20.000000
mean	43000.000000	664.000000	223250.000000
std	17923.815383	85.526235	83733.207272
min	22000.000000	555.000000	120000.000000
25%	26750.000000	583.750000	150000.000000
50%	40000.000000	660.000000	200000.000000
75%	56250.000000	735.000000	277500.000000
max	75000.000000	790.000000	400000.000000





5. Agriculture Crop Yield Dataset

Question 5

An agricultural department wants to analyze crop yield based on rainfall, temperature, fertilizer usage, soil type, and crop yield. Perform EDA to identify the factors influencing crop production.

Farm_ID	Rainfall_mm	Temperature	Fertilizer_kg	Soil_Type	Crop_Yield_kg
F01	850	28	60	Loamy	3200
F02	600	32	45	Sandy	2100
F03	900	27	65	Loamy	3500
F04	500	34	40	Sandy	1800
F05	780	29	55	Clay	3000
F06	650	31	48	Sandy	2300
F07	920	26	70	Loamy	3700
F08	720	30	52	Clay	2800
F09	550	33	42	Sandy	1900

F10	800	28	58	Loamy	3100
F11	880	27	62	Loamy	3400
F12	620	32	46	Sandy	2200
F13	760	29	54	Clay	2950
F14	940	26	72	Loamy	3800
F15	580	33	44	Sandy	2000
F16	820	28	60	Clay	3150
F17	700	30	50	Clay	2700
F18	960	25	75	Loamy	3900
F19	540	34	41	Sandy	1850
F20	790	29	56	Clay	3050

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# Load the Agriculture Crop Yield dataset
file_path_agriculture = '/content/Agriculture_Crop_Yield.xlsx'
df_agriculture = pd.read_excel(file_path_agriculture)

# Display the first 5 rows of the DataFrame
print('First 5 rows of the Agriculture Crop Yield dataset:')
display(df_agriculture.head())
# Display basic information about the agriculture dataset
print('\nInformation about the Agriculture Crop Yield dataset:')
df_agriculture.info()
# Display descriptive statistics for the agriculture dataset
print('\nDescriptive statistics of the Agriculture Crop Yield dataset:')
display(df_agriculture.describe())
# Check for missing values in the agriculture dataset
print('\nMissing values in each column of the Agriculture Crop Yield dataset:')
display(df_agriculture.isnull().sum())
numerical_features_agriculture = ['Rainfall_mm', 'Temperature', 'Fertilizer_kg',
'Crop_Yield_kg']

plt.figure(figsize=(15, 10))
for i, feature in enumerate(numerical_features_agriculture):
    plt.subplot(2, 2, i + 1)
    sns.histplot(df_agriculture[feature], kde=True)
    plt.title(f'Distribution of {feature}')
    plt.xlabel(feature)
    plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
plt.figure(figsize=(18, 6))

# Bar plot for average Crop Yield by Rainfall (categorized if needed, or direct if discrete)
sns.barplot(x='Rainfall_mm', y='Crop_Yield_kg', data=df_agriculture, palette='viridis')
plt.title('Average Crop Yield by Rainfall')
plt.xlabel('Rainfall (mm)')
```

```
plt.ylabel('Average Crop Yield (kg)')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

```
plt.figure(figsize=(18, 6))
# Bar plot for average Crop Yield by Temperature
sns.barplot(x='Temperature', y='Crop_Yield_kg', data=df_agriculture, palette='plasma')
plt.title('Average Crop Yield by Temperature')
plt.xlabel('Temperature')
plt.ylabel('Average Crop Yield (kg)')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

```
plt.figure(figsize=(18, 6))
# Bar plot for average Crop Yield by Fertilizer Usage
sns.barplot(x='Fertilizer_kg', y='Crop_Yield_kg', data=df_agriculture, palette='magma')
plt.title('Average Crop Yield by Fertilizer Usage')
plt.xlabel('Fertilizer Usage (kg)')
plt.ylabel('Average Crop Yield (kg)')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

```
plt.figure(figsize=(10, 6))
# Bar plot for average Crop Yield by Soil Type
sns.barplot(x='Soil_Type', y='Crop_Yield_kg', data=df_agriculture, palette='cividis')
plt.title('Average Crop Yield by Soil Type')
plt.xlabel('Soil Type')
plt.ylabel('Average Crop Yield (kg)')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

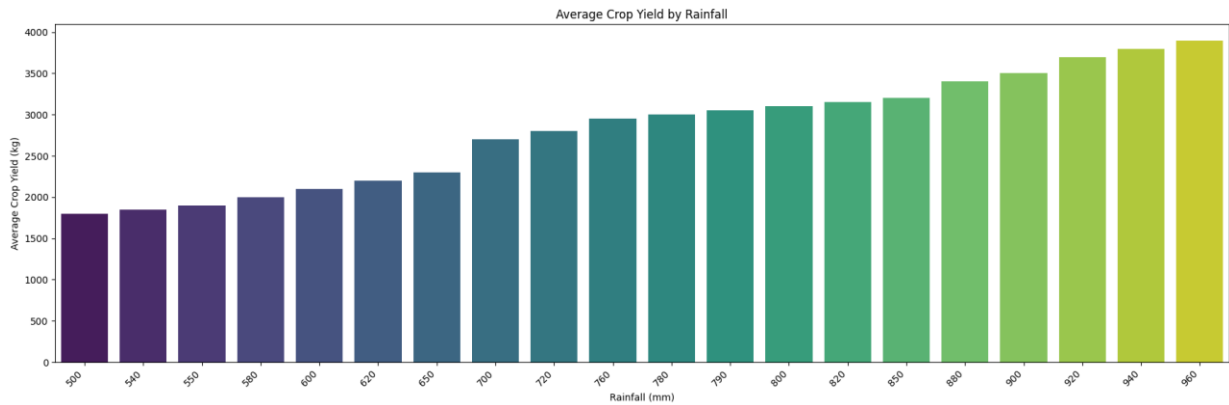
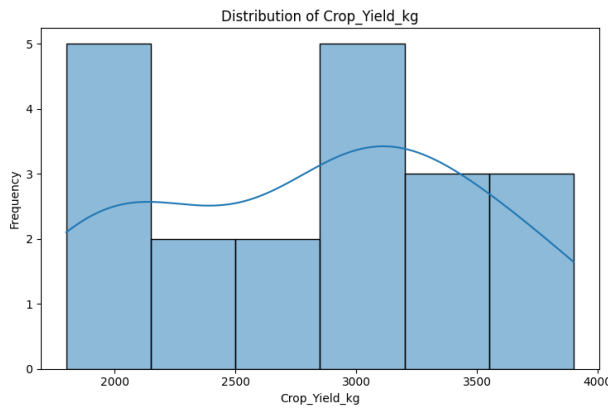
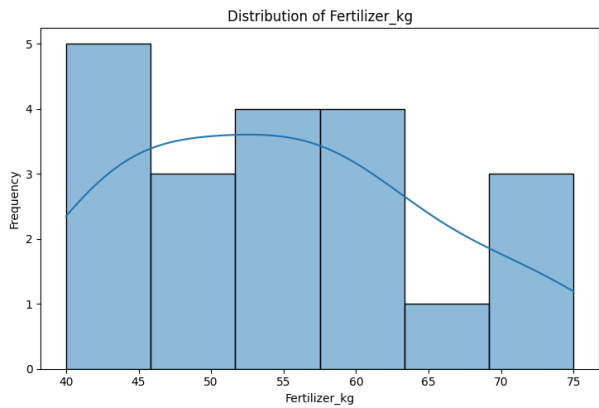
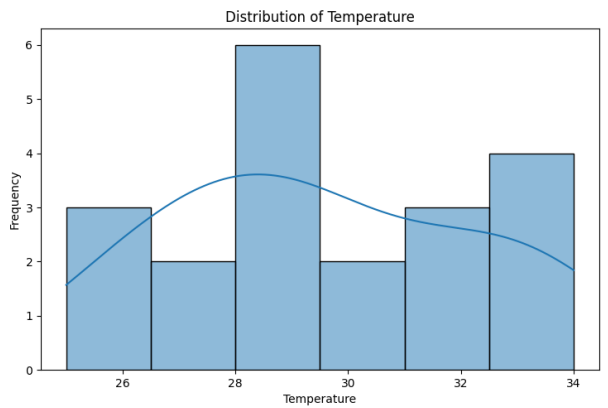
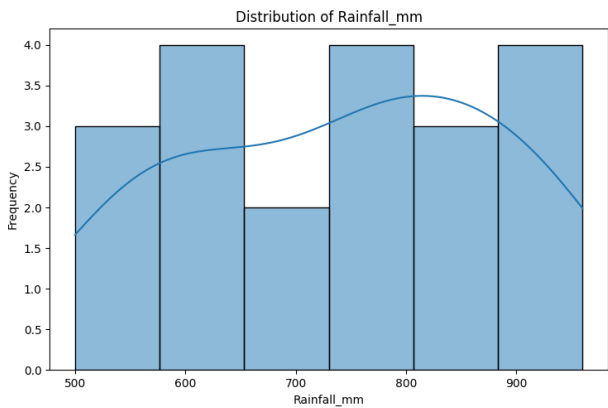
ANSWER:

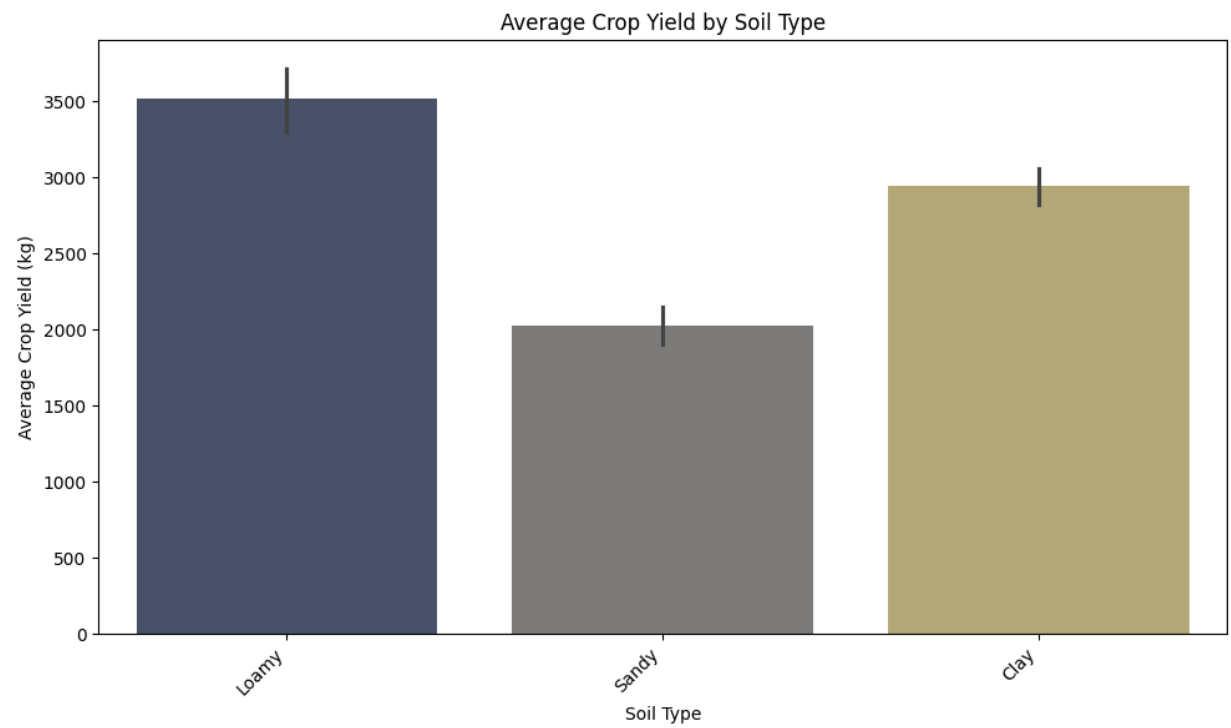
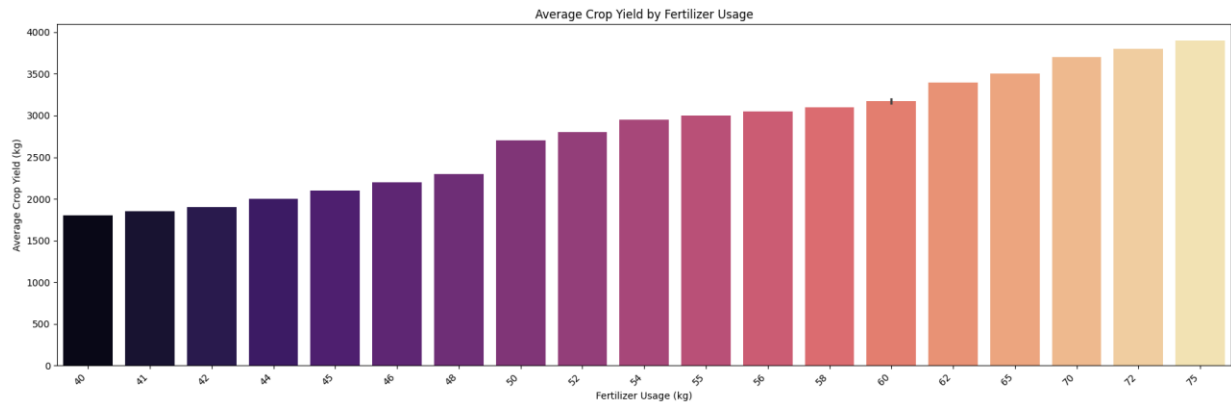
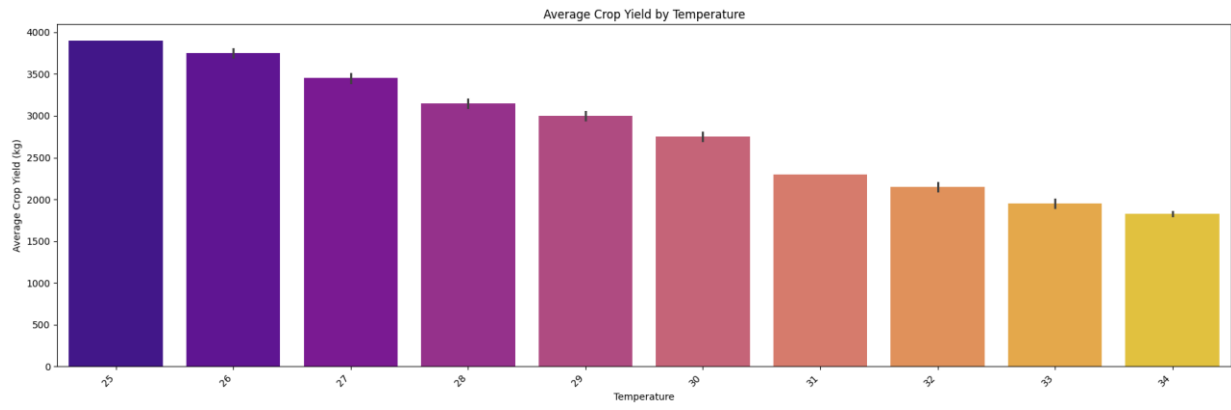
First 5 rows of the Agriculture Crop Yield dataset:

	Farm_ID	Rainfall_mm	Temperature	Fertilizer_kg	Soil_Type	Crop_Yield_kg
0	F01	850	28	60	Loamy	3200
1	F02	600	32	45	Sandy	2100
2	F03	900	27	65	Loamy	3500
3	F04	500	34	40	Sandy	1800
4	F05	780	29	55	Clay	3000

Descriptive statistics of the Agriculture Crop Yield dataset:

	Rainfall_mm	Temperature	Fertilizer_kg	Crop_Yield_kg
count	20.000000	20.000000	20.000000	20.000000
mean	743.000000	29.550000	54.750000	2820.000000
std	144.335136	2.762055	10.507516	681.407213
min	500.000000	25.000000	40.000000	1800.000000
25%	615.000000	27.750000	45.750000	2175.000000
50%	770.000000	29.000000	54.500000	2975.000000
75%	857.500000	32.000000	60.500000	3250.000000
max	960.000000	34.000000	75.000000	3900.000000





Common EDA Tasks:

1. Identify the number of rows and columns.
2. Check data types of each column.
3. Find missing values and duplicate values.
4. Calculate mean, median, minimum, maximum, and standard deviation.
5. Draw suitable charts such as bar chart, histogram, scatter plot, and box plot.
6. Identify important patterns from the dataset.
7. Write 4 to 5 observations based on EDA.
8. Give a short conclusion from the analysis.

RUBRICS FOR CALCULATION

Exploratory Data Analysis-Based Practical Assessment					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Dataset Understanding	20	Clearly explains dataset source, features, target variables, data types, and problem context with strong clarity	Explains dataset, features, and variables with adequate clarity	Basic explanation of dataset with limited understanding of variables	Poor understanding of dataset; variables and problem context are unclear
Data Cleaning	25	Effectively identifies and handles missing values, duplicates, incorrect entries, and outliers using suitable methods	Handles most missing values, duplicates, and errors correctly	Performs basic cleaning but misses some important data quality issues	Minimal or incorrect data cleaning; major errors remain in the dataset
Data Analysis and Statistical Summary	20	Provides detailed descriptive statistics, comparisons, patterns, and meaningful interpretations	Provides relevant summary statistics and basic interpretation	Provides limited statistics with weak explanation	Lacks statistical analysis or gives incorrect interpretation
Data Visualization	20	Uses highly appropriate charts with proper labels, titles, and clear visual interpretation	Uses suitable charts with mostly clear labels and explanation	Uses limited or basic charts with unclear interpretation	Poor or incorrect visualizations; charts do not support analysis
Insight Generation and Reporting	15	Presents strong insights, conclusions, and recommendations based on data evidence	Presents clear findings and conclusions with reasonable support	Provides basic findings but lacks depth or proper justification	Findings are unclear, unsupported, or not related to the data

CO1-ASSESSMENT TOOL -3 Exploratory Data Analysis

QUESTIONS : 5

MARKS : 50

Q No	Dataset Understanding (20%)	Data Cleaning (25%)	Data Analysis and Statistical Summary (20%)	Data Visualization (20%)	Insight Generation and Reporting (15%)	Total Score (100%)
1	/2.5	/2.5	/2.0	/2.0	/1.0	/10
2	/2.5	/2.5	/2.0	/2.0	/1.0	/10
3	/2.5	/2.5	/2.0	/2.0	/1.0	/10
4	/2.5	/2.5	/2.0	/2.0	/1.0	/10
5	/2.5	/2.5	/2.0	/2.0	/1.0	/10
Overall Rubrics Value	/12.5	/12.5	/10	/10	/5	/50
	/25	/25	/20	/20	/10	/100